HK IMO Training 2021-2022
Problem Set 24 Suggested Solutions

Problem 93. Let n be a positive integer. In how many ways can we fill in the numbers $1, 2, \ldots, 3n$ into the cells of a $3 \times n$ table such that the n numbers in each row are in ascending order (from left to right), and the 3 numbers in each column are in ascending order (from top to bottom)?

Solution. For the cell (i, j) in the ith row and the jth column, there are $(3-i)+(n-j)+1 = n-i-j+4$ cells (i', j') satisfying $i' \geqslant i$ and $j' = j$, or $j' \geqslant j$ and $i' = i$. By the hook length formula, the number of ways to fill in the table is

$$\frac{(3n)!}{\prod\limits_{\substack{1 \leqslant i \leqslant 3 \\ 1 \leqslant j \leqslant n}} (n-i-j+4)} = \frac{2 \cdot (3n)!}{(n+2)!(n+1)!n!}.$$

$\square$

Remark. Indeed, the correct formulation of the problem is as follows (and the answer is $\dfrac{(3n)!}{n! \cdot 6^n}$):

Problem (Norway MO Final 2021 Problem 1a)**.** Let n be a positive integer. In how many ways can we fill in the numbers $1, 2, \ldots, 3n$ into the cells of a $3 \times n$ table such that the n numbers in the first row are in ascending order (from left to right), and the 3 numbers in each column are in ascending order (from top to bottom)?

Problem 94 (Norway MO Final 2016 Problem 4)**.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x+y)f(y) = |x|\, f\left(y + \frac{y^2+1}{x}\right) \tag{1}$$

for any real numbers x and y with $x \neq 0$.

Solution. The solutions are the zero function and the function $f(x) = \sqrt{x^2+1}$. The former is clearly a solution. For the latter, we check that

$$f(x+y)f(y) = \sqrt{(x+y)^2+1}\sqrt{y^2+1} = \sqrt{[(x+y)^2+1](y^2+1)},$$
$$|x|\, f\left(y + \frac{y^2+1}{x}\right) = |x| \sqrt{\left(y + \frac{y^2+1}{x}\right)^2 + 1} = \sqrt{(xy+y^2+1)^2 + x^2}.$$

It is routine to check that the two expressions are the same.

We now prove that these are the only solutions. If $f(t) = 0$ for some $t \in \mathbb{R}$, by putting $y = t$ in (1), we obtain $f\left(t + \dfrac{t^2+1}{x}\right) = 0$ for any $x \neq 0$. Note that $t + \dfrac{t^2+1}{x}$ can be any real value different from t. Thus, $f(z) = 0$ for any $z \neq t$, and so f is the zero function.

If $f(x) \neq 0$ for all $x \in \mathbb{R}$, we choose $x = \sqrt{y^2+1}$ in (1). Note that $y + \dfrac{y^2+1}{x} = y + \sqrt{y^2+1} = y + x$. As $f(x+y) \neq 0$, equation (1) implies $f(y) = \left|\sqrt{y^2+1}\right| = \sqrt{y^2+1}$. This shows there are only two solutions. $\square$

Problem 95 (Iran TST 2017 Test 3 Problem 1)**.** Let $n > 1$ be an integer. Prove that there exists an integer m with $\left\lfloor \dfrac{n}{2} \right\rfloor \leqslant m \leqslant n-1$, and integers $a_1, a_2, \ldots, a_{n-m}$ such that $a_1 > 0$ and

$$\frac{a_1}{m+1} + \frac{a_2}{m+2} + \cdots + \frac{a_{n-m}}{n} = \frac{1}{\operatorname{lcm}(1, 2, \ldots, n)}.$$

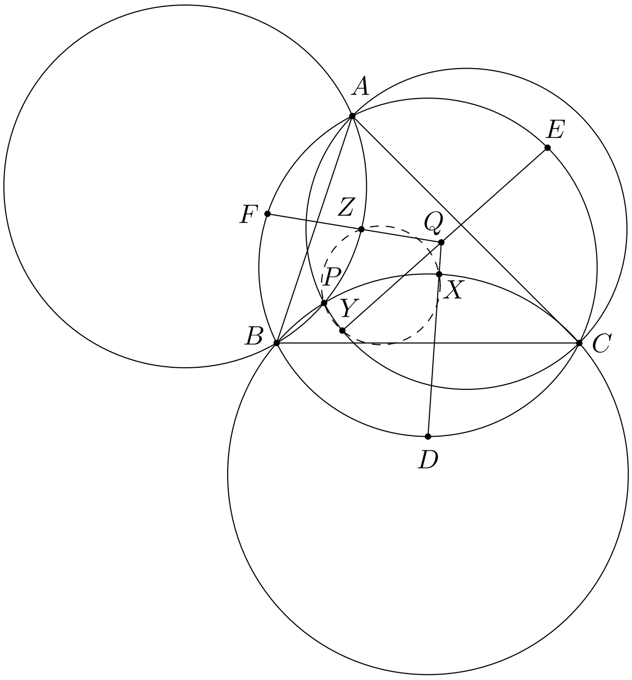
Solution. Let $m = \left\lfloor \dfrac{n}{2} \right\rfloor$. For each $k = 1, 2, \ldots, m$, there is a multiple of k among $m+1, m+2, \ldots, n$ since there are $n-m \geqslant m \geqslant k$ consecutive integers. Therefore, we have $L := \operatorname{lcm}(1, 2, \ldots, n) = \operatorname{lcm}(m+1, m+2, \ldots, n)$.

Now, since $\gcd\left(\frac{L}{m+1}, \frac{L}{m+2}, \ldots, \frac{L}{n}\right) = 1$, there exists $a_1, a_2, \ldots, a_{n-m}$ such that

$$\frac{L}{m+1}a_1 + \frac{L}{m+2}a_2 + \cdots + \frac{L}{n}a_{n-m} = 1.$$

Replacing a_1 and a_2 by $a_1 + \frac{L}{m+2}t$ and $a_2 - \frac{L}{m+1}t$ respectively for a sufficiently large t if necessary, we may assume $a_1 > 0$. This yields the desired result readily. $\square$

Problem 96 (KöMaL Points Contest A. 754). Let P be a point inside an acute $\triangle ABC$, and let Q be the isogonal conjugate of P with respect to $\triangle ABC$. Let D, E, F be the midpoints of the minor arcs BC, CA, AB of the circumcircle of $\triangle ABC$ respectively. The rays DQ, EQ, FQ meet the circumcircles of $\triangle BCP$, $\triangle CAP$, $\triangle ABP$ at X, Y, Z respectively. Prove that P, X, Y, Z are concyclic.



Solution. Let W be the point on the ray DQ such that $DW \times DQ = DB^2 = DC^2$. We claim that $W = X$. Indeed, since $\triangle DBW \sim \triangle DQB$ and $\triangle DCW \sim \triangle DQC$, we have

$$\begin{aligned}\angle BWC = \angle BWD + \angle CWD = \angle QBD + \angle QCD = 360^\circ - \angle BQC - \angle BDC = (180^\circ - \angle BQC) + \angle BAC \\ = \angle QBC + \angle QCB + \angle BAC = 180^\circ - \angle ABQ - \angle ACQ = 180^\circ - \angle CBP - \angle BCP = \angle BPC.\end{aligned}$$

This proves W lies on (BPC), and hence $W = X$. Therefore, we obtain $\frac{BX}{BQ} = \frac{DX}{BD} = \frac{DX}{CD} = \frac{CX}{CQ}$, and hence $\frac{BX}{CX} = \frac{BQ}{CQ}$. Similarly, we have $\frac{CY}{AY} = \frac{CQ}{AQ}$ and $\frac{AZ}{BZ} = \frac{AQ}{BQ}$.

Consider the inversion with centre P and radius 1. Let A' be the image of A, etc. Note that X', Y', Z' lie on the extensions of the lines $B'C', C'A', A'B'$ respectively (and not lying on the segments $B'C', C'A', A'B'$ respectively). Since

$$\frac{B'X'}{X'C'} = \frac{\frac{1}{PB' \times PX'}BX}{\frac{1}{PC' \times PX'}CX} = \frac{PC'}{PB'} \cdot \frac{BX}{CX},$$

we have

$$\begin{aligned}\frac{B'X'}{X'C'} \times \frac{C'Y'}{Y'A'} \times \frac{A'Z'}{Z'B'} &= \left(\frac{PC'}{PB'} \times \frac{PA'}{PC'} \times \frac{PB'}{PA'}\right)\left(\frac{BX}{CX} \times \frac{CY}{AY} \times \frac{AZ}{BZ}\right) \\ &= \frac{BQ}{CQ} \times \frac{CQ}{AQ} \times \frac{AQ}{BQ} = 1.\end{aligned}$$

By Menelaus' theorem, the points X', Y', Z' are collinear. This means X, Y, Z, P are concyclic. $\square$